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Introduction to Blockchain Technology

Lesson 3 - Cryptography Basis of Blockchain



Cryptography Basis of Blockchain



- 1. Why Cryptography is Needed in Blockchain**
- 2. Cryptography Basics**
- 3. Hashing in Blockchain**
- 4. Public Key Cryptography**
- 5. Digital Signatures**
- 6. Cryptography in Blockchain Transactions**
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1. Why Cryptography is Needed in Blockchain

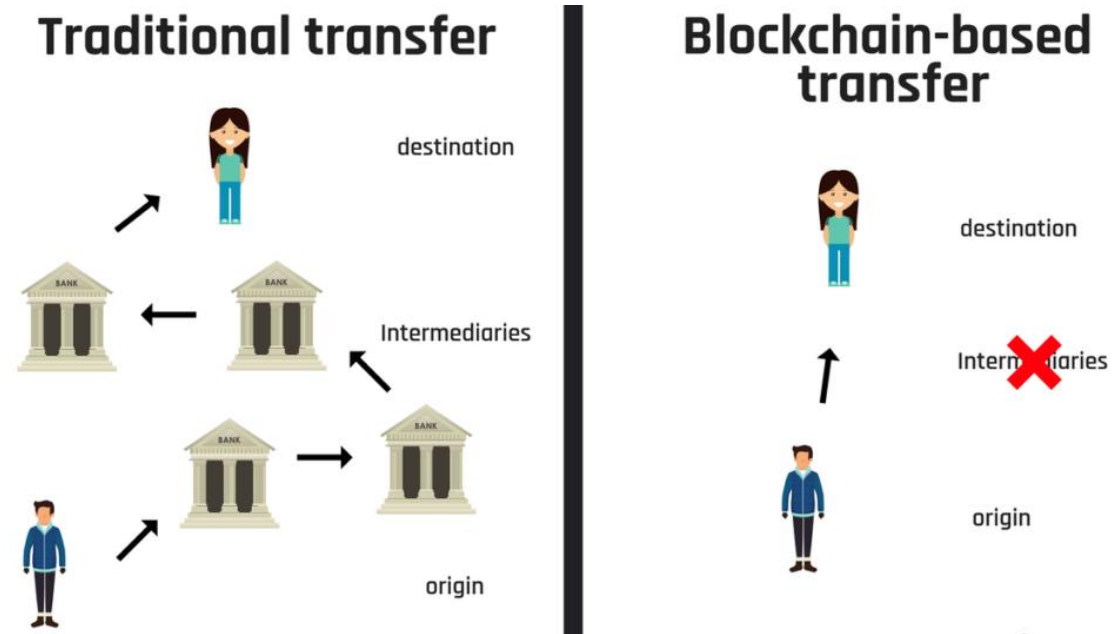
- ❖ Decentralized
- ❖ Distributed
- ❖ Publicly verifiable

Security challenges

- ❖ No central authority
- ❖ Data shared across nodes
- ❖ Open participation

=> Cryptography ensures **trust, security, and integrity** without a trusted third party.

Q: Can traditional database security mechanisms (Admin, Access Control, etc.) be used in blockchain?



1. Why Cryptography is Needed in Blockchain

Cryptography and the CIA Triad in Blockchain

- ❖ Confidentiality (C): Ensuring only authorized users can access data. In public blockchains, this is often handled off-chain or via zero-knowledge proofs
- ❖ Integrity (I): Guaranteeing that data has not been modified. This is achieved through one-way hash functions. If the computed hash does not match the received data, the system detects a breach
- ❖ Availability (A): Ensuring data and tools are accessible to all authorized applications. Blockchain achieves this through redundancy across P2P nodes, preventing a "single point of failure"

=> Cryptography in blockchain moves from **trusting people** to **trusting mathematical proof**

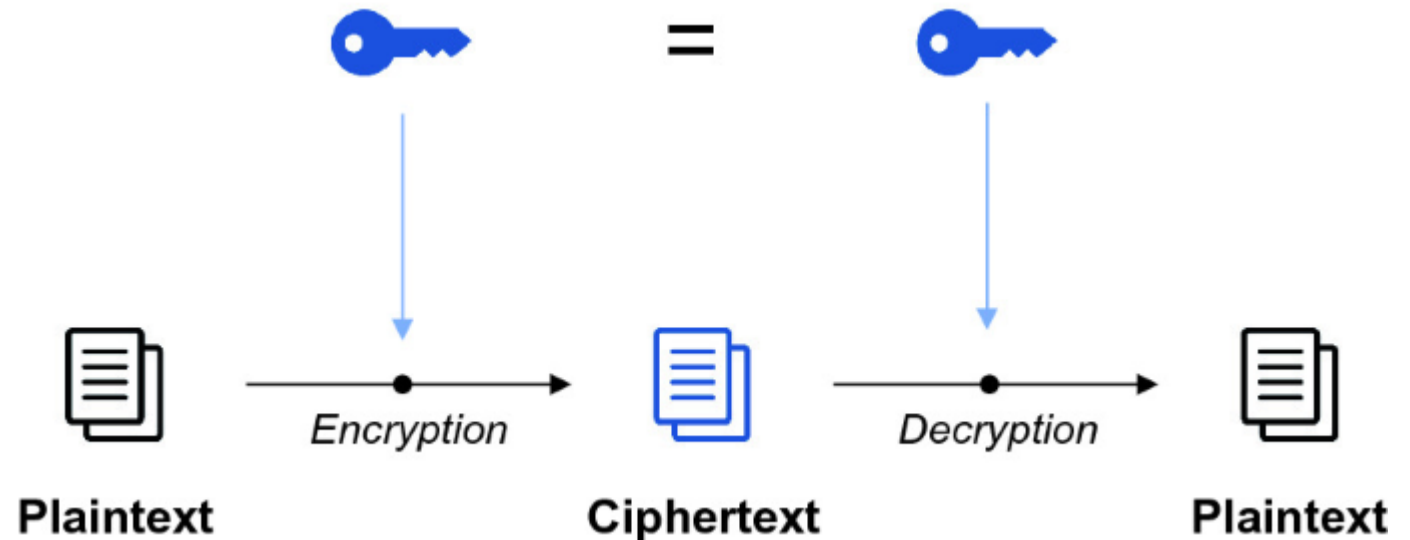
2. Cryptography Basics

Cryptography

- ❖ Technique to secure data and communication
- ❖ Based on mathematical algorithms

The goal of using Cryptography:

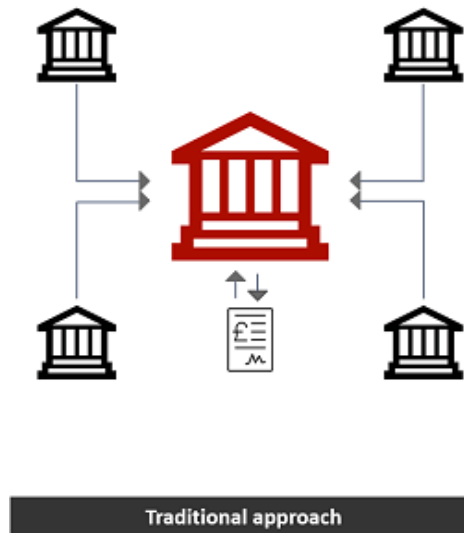
- ❖ Confidentiality
- ❖ **Integrity**
- ❖ **Authentication**
- ❖ Non-repudiation



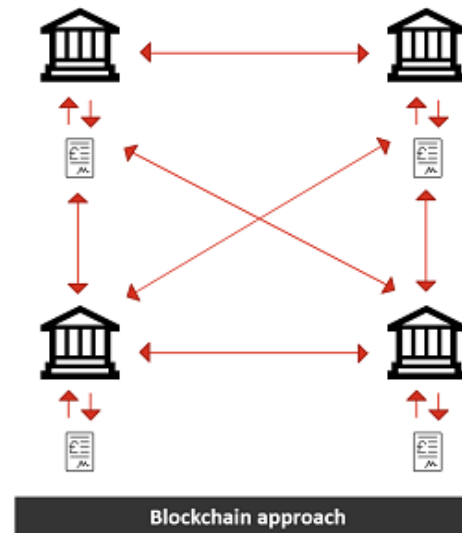
2. Cryptography Basics

Cryptography in Traditional Systems vs Blockchain

Traditional IT Systems	Blockchain
Central authentication server	No central authority
Username + password	Private key + signature
Admin-controlled access	Cryptographic verification
Trust-based	Math-based



Database is controlled by a central and trusted third-party



Each participant has a copy of the database, ensuring immutability

2. Cryptography Basics

Blockchains mainly use **Encryption** and **Hashing**.

Method	Reversible	Purpose
Encoding	Yes	Data format
Encryption	Yes (with key)	Confidentiality
Hashing	No	Data integrity

Q: Why **hashing** is needed in blockchain systems, besides **encryption**?

3. Hashing in Blockchain

A hash function:

- ❖ Takes input of any size
- ❖ Produces fixed-length output

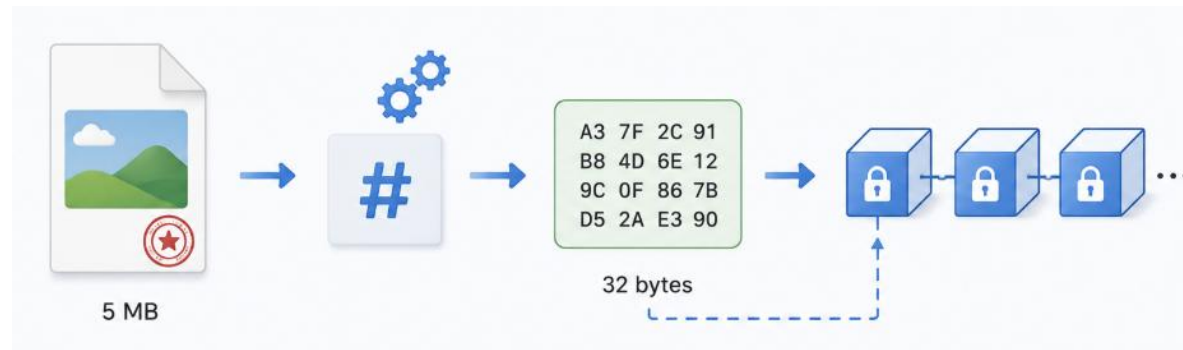
Properties of Secure Hash Functions

- ❖ **Deterministic:** The same input always produces the same hash output.
- ❖ **Fast computation:** Hash values must be computed efficiently, even for large inputs.
- ❖ **Pre-image resistance:** Given a hash output, it should be computationally infeasible to find the original input.
- ❖ **Collision resistance:** It should be extremely difficult to find two different inputs that produce the same hash.
- ❖ **Avalanche effect:** A small change in input causes a completely different hash output.

Q: Which hash property ensures that modifying one transaction invalidates all following blocks?

3. Hashing in Blockchain

Example: Hashing a 5MB image of a land deed results in a 32-byte string. Storing this on-chain allows the system to verify the 5MB file's integrity without storing the heavy image itself
Blockchain stores the fingerprint of the file, not the file itself.



Verification



3. Hashing in Blockchain

Hashing is used to:

- ❖ Identify transactions
- ❖ Link blocks together
- ❖ Build Merkle Trees
- ❖ Protect historical data

Block linking

- ❖ Each block stores the hash of the previous block

Q: What happens if an attacker changes a transaction in an old block?

4. Public Key Cryptography

Aspect	Symmetric Cryptography	Asymmetric Cryptography
Number of keys	One shared secret key	Two keys (public & private)
Key usage	Same key encrypts and decrypts	Public key encrypts / verifies, private key decrypts / signs
Key distribution	Requires secure key sharing	Public key can be shared openly
Performance	Fast, low computational cost	Slower than symmetric
Scalability	Poor (key sharing grows with users)	Good (no shared secret needed)
Security risk	Key leakage compromises all communication	Private key never shared
Examples	AES, DES	RSA, ECC

Q: Which one is more suitable for decentralized systems?

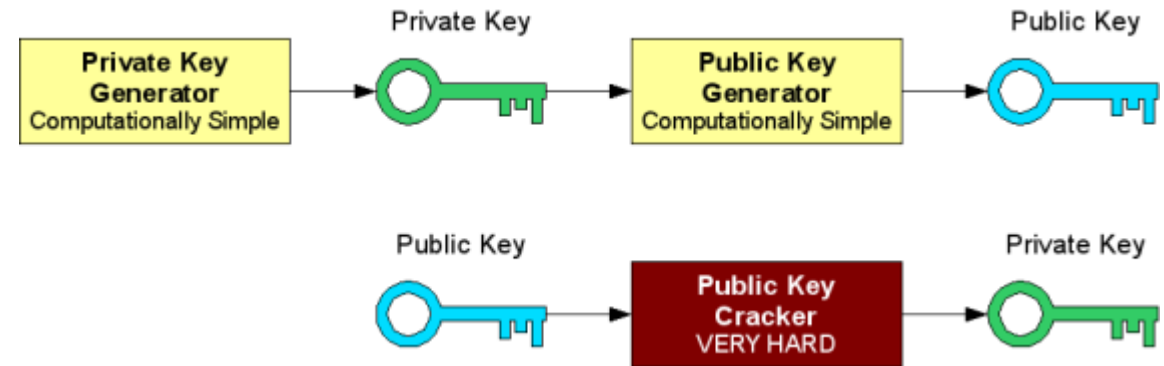
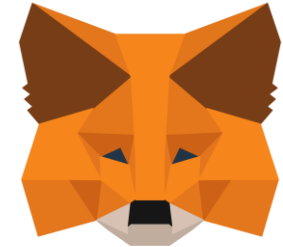
4. Public Key Cryptography

Blockchain uses **asymmetric cryptography** to manage identity and ownership in a decentralized network.

Each user controls a **key pair**:

- ❖ **Private key**: kept secret
- ❖ **Public key**: shared with the network

Blockchain Wallet



Key characteristics

- ❖ Public key is mathematically derived from private key
- ❖ It is computationally infeasible to derive the private key from the public key

Q: Why use **asymmetric** cryptography instead of **symmetric** cryptography in blockchain?

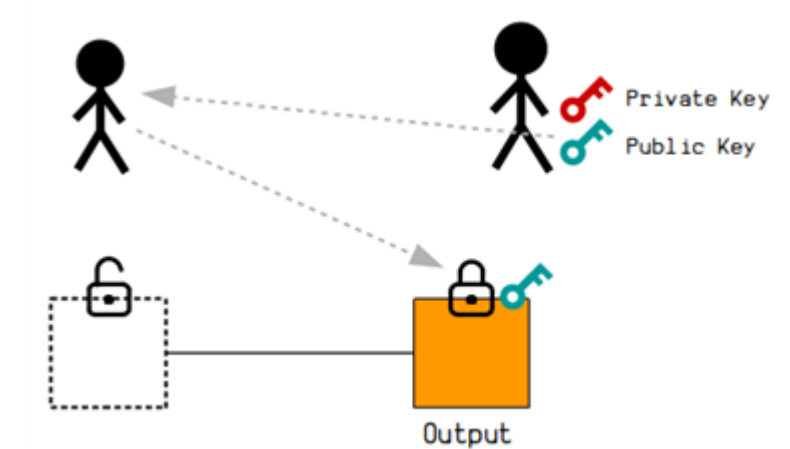
4. Public Key Cryptography

Private Key: the most critical security element in blockchain systems

- ❖ A large random number (e.g., 256-bit)
- ❖ Used to sign transactions
- ❖ Proves ownership of blockchain assets

Important

- ❖ Loss of private key = permanent loss of assets



Example: Alice owns Bitcoin. Alice signs the transaction using her private key. The network accepts the transaction only if the signature is valid.

Q: Why is private key management critical in blockchain systems?

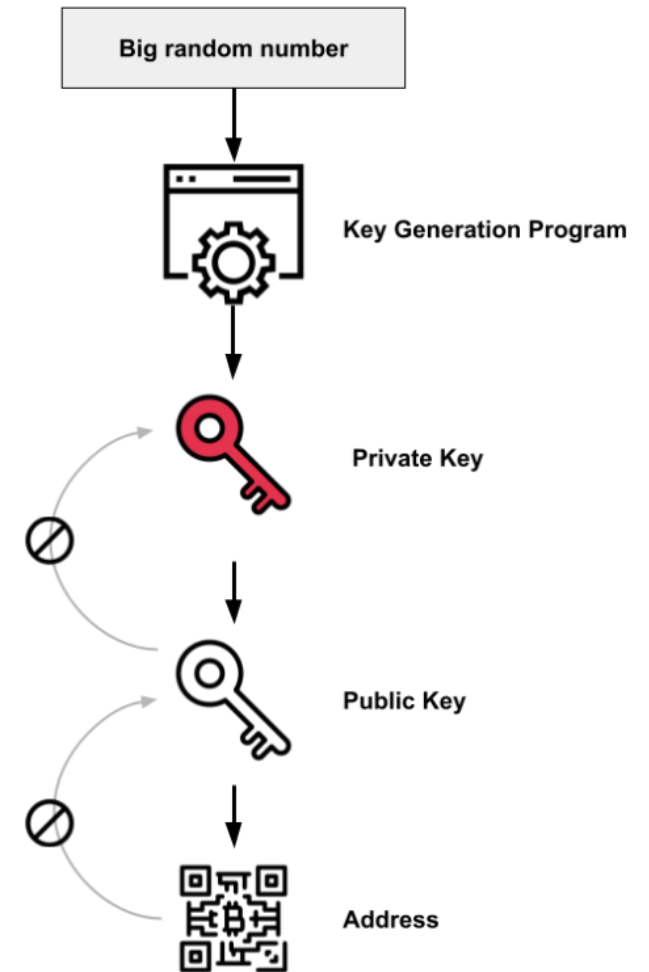
4. Public Key Cryptography

Public Key

- ❖ Derived from private key
- ❖ Used to verify digital signatures

Blockchain Address

- ❖ A shortened, hashed version of the public key
- ❖ Used as sender and receiver identifier



Q: Why does blockchain use addresses instead of public keys directly?

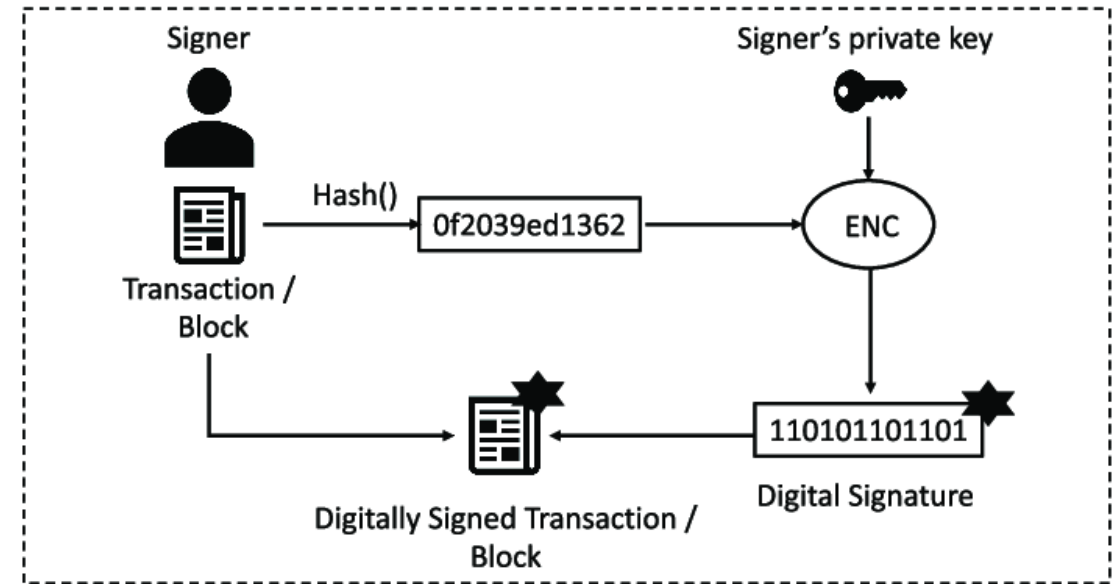
5. Digital Signatures

A **digital signature** proves:

- ❖ Who created the transaction
- ❖ That the transaction was not altered

Signing process

- 1) Transaction data is hashed
- 2) Hash is encrypted using the sender's private key
- 3) Signature is attached to the transaction



Q: Why is the transaction hashed before signing instead of signing raw data?

5. Digital Signatures

Signature Verification

- ❖ Network receives transaction + signature
- ❖ Hash transaction data
- ❖ Use sender's public key to verify signature

If verification fails

- ❖ Transaction is rejected immediately

Security guarantees

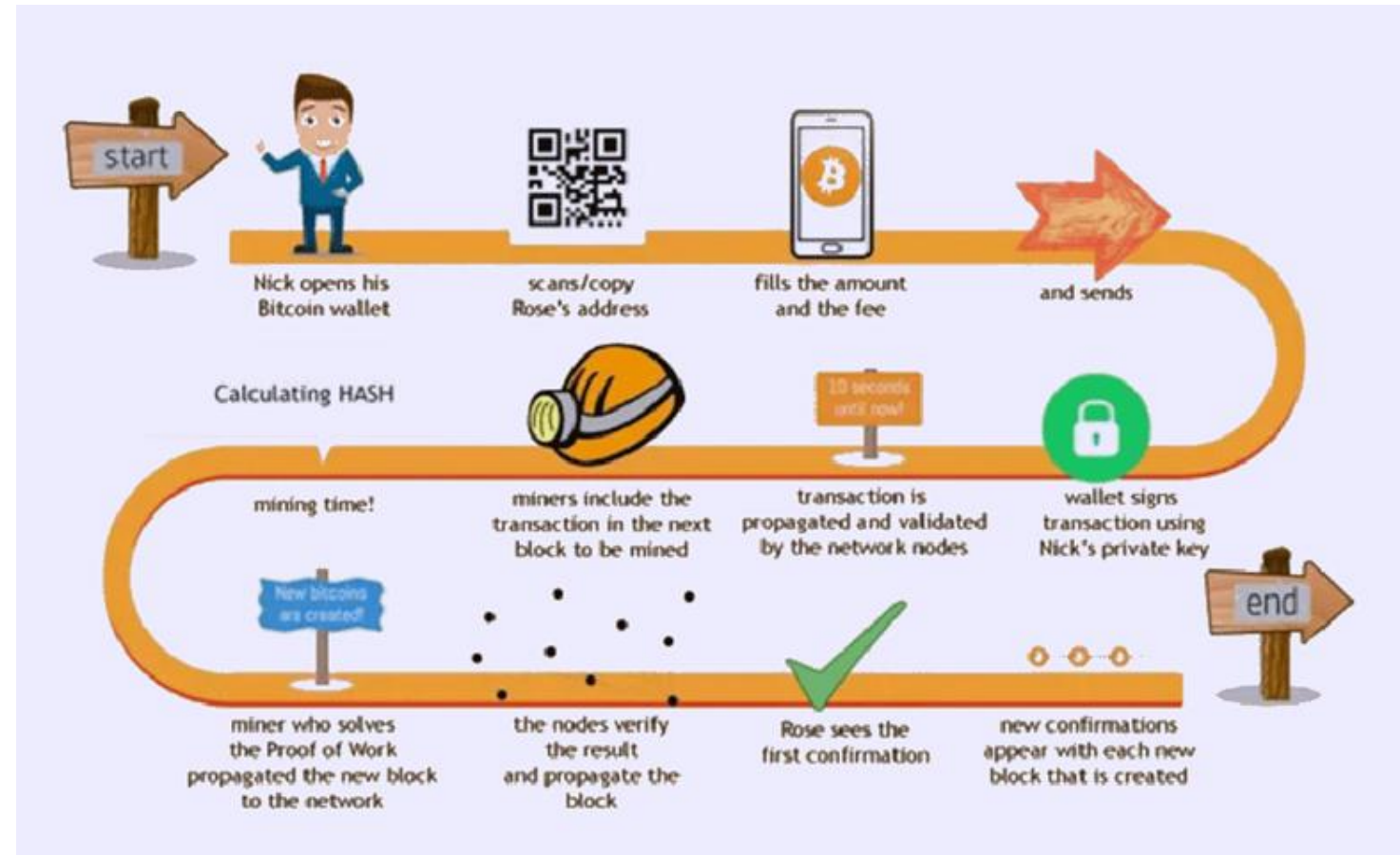
- ❖ Authentication (who sent it)
- ❖ Integrity (data unchanged)
- ❖ Non-repudiation (sender cannot deny it)

Q: Can someone modify a signed transaction without invalidating it?

6. Cryptography in Blockchain Transactions

Transaction security workflow

- 1) User creates transaction
- 2) Transaction is hashed
- 3) Hash is digitally signed
- 4) Transaction broadcast to network
- 5) Nodes verify signature
- 6) Valid transaction added to block



6. Cryptography in Blockchain Transactions

Cryptographic Components Summary

Cryptographic Tool	Purpose
Hash function	Data integrity
Private key	Ownership
Public key	Verification
Address	Identity
Digital signature	Transaction security

=> Blockchain security depends on mathematics, not trust in institutions.

7. Summary

- ❖ Hash functions protect data integrity
- ❖ Private keys represent ownership
- ❖ Public keys enable verification
- ❖ Digital signatures secure transactions
- ❖ Cryptography replaces centralized trust mechanisms

8. Discussion

1. Why is private key management considered the weakest point in blockchain security?
2. How do digital signatures replace usernames and passwords in blockchain systems?
3. Blockchains rely on secure hash functions (e.g., SHA-256). Which hash properties are most important for preventing data tampering?
4. What would happen to a blockchain system if collision resistance were broken?
5. Why must every node verify digital signatures independently?
6. Can a user change their Private Key if it is compromised?